



Research paper

TransIST: Transformer based infrared small target tracking using multi-scale feature and exponential moving average learning

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ARTICLE INFO

Keywords:

Infrared tracking

Small objects

Transformer

Multi-scale dilated attention

Exponential moving average

Edge suppression

ABSTRACT

Small Unmanned Aerial Vehicle (UAV) tracking against complex sky backgrounds is of significant importance in both the military and civilian domains, with traditional correlation filters imposing high requirements on feature models. Consequently, a deep learning model is designed to achieve effective infrared tracking of a small UAV target. Specifically, a transformer model is used as the backbone, and a multi-scale attention is introduced to obtain the perceptual features referring to small targets. Then, an edge suppression model named side window filter is embedded to suppress the negative effect of edge on small target tracking. Furthermore, the proposed model is trained using an exponential moving average learning strategy, which results in more precise network parameters. Experimental results demonstrate that the proposed Transformer-based Infrared Small Target Tracking (TransIST) algorithm exhibits superior performance in public near-infrared videos compared to current related algorithms, with enhanced stability over correlation filtering-based trackers. The code will be available at <https://github.com/len-ayan/TransIST>, contributing to the remote sensing community.

1. Introduction

Near Infrared Radiation (NIR) is essential to identify an Unmanned Aerial Vehicle (UAV). However, tracking small targets involves identifying objects that have reduced thermal features. In addition, the diminutive size of an object, coupled with a complex background and the presence of false targets that resemble it, creates challenges in reliably assessing the targets' status. To address these obstacles, it is crucial to develop algorithms that can precisely identify small targets in complicated environments [1].

Traditional small target tracking methods contain mainly Detect Before the Track (DBT) [2,3] and Track Before Detect (TBD) [4–6]. Emphasizing the early detection and identification of potential threats or targets is central to the DBT method, which occurs prior to initiating tracking. However, this approach does not gather information about the target due to the presence of false targets. In the TBD method, the tracking of an object begins prior to its detection. This technique is frequently employed when the target has already been identified, allowing for the continuous monitoring of its motion. Practically speaking, the focus of the DBT method is on enhancing the signal-to-noise ratio by leveraging the accumulated energy in the target path [6]. Nevertheless,

conducting a thorough search for targets is a time-intensive process. To summarize, both types of algorithms encounter challenges in detection and recognition [7,8] due to significant background noise and faint targets.

Progress in Infrared Small target Tracking (IST) has been significantly influenced by key advances in the domain of object tracking, which contains Discriminative Correlation Filters (DCF) [1,9–12] and Deep Learning (DL). The Kernelized Correlation Filter (KCF) [11] performs well in feature extraction and benefits from the kernel trick to improve speed. For adaptation in IST, Qian et al. proposed an improved KCF algorithm whose performance can be affected by boundary issues. Next, the singular value decomposition method for background suppression is integrated with the KCF model [13]. However, the algorithm's sensitivity to complex backgrounds arises because of the necessity for manual parameter tuning. To reduce the impact of edge effects, guided image filtering is utilized to enhance the results of the KCF tracker [13]. Note that the guided filter needs to obtain the guide image in advance. To conclude, while these DCF techniques have successfully overcome various obstacles in target tracking, the scarcity of data complicates the tracking of infrared targets [14,15]. Additional techniques such as sparse representation [16] and particle

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filtering [17] also exist. Nevertheless, achieving real-time performance is an essential factor in object tracking. Additionally, the Spatial-Temporal Regularized Correlation Filter (STRCF) [18,19] has added regularization terms to prevent model deterioration. To optimize the continuity of small target tracking, STRCF is combined with the detection model (named STRCFD) using an adaptive search factor [20]. The aforementioned conventional trackers exhibit poor performance in feature classification, an issue that has been addressed by DL models [21–25]. The SiamIST [25] model, which is a variation of SiamRPN (Siamese Region Proposal Network), effectively identifies patterns in infrared images. However, its performance is limited due to a complex anchor mechanism. In recent times, the widely-used Transformer Tracking (TransT) has demonstrated substantial enhancements in tracking capability, owing to a feature fusion network consisting of the Cross-Feature Augment (CFA) model and the Ego-Context Augment (ECA) model. Additionally, the ET-tracker [26] enhances the tracking network by utilizing instance-level attention. Tomp [27] employs an innovative transformer architecture to utilize prior information for target tracking. Nonetheless, these DL models are not suitable for tracking small targets in the absence of training images. It should be mentioned that several labeled infrared data sets that involve detection and tracking algorithms [28–32] have been made publicly available, which offers the possibility for DL models to track small targets.

In conclusion, an IST algorithm is constructed based on TransT using the public data set of small targets [33]. A deep learning model is developed to handle variations in the appearance of the target and complex environment. The architecture utilizes the transformer model as its core, integrating a specially designed Multi-Scale Dilated Attention (MSDA) to improve the feature extraction model's capabilities. Additionally, the use of a Side Window Filter (SWF) mitigates the effect of image edges, and the Exponential Moving Average (EMA) techniques enhance training stability to achieve improved tracking performance. Research indicates that the proposed tracking algorithm achieves accurate tracking results in challenging environments. The principal contributions are outlined below.

- A small UAV tracking method on a public NIR data set is developed using transformer-based technology for the first time.
- The transformer framework is enhanced with the MSDA, considerably enhancing the tracking accuracy for small objects. Moreover, the SWF method helps to reduce distortions caused by edge.
- The use of the EMA technique allows for the smoothing of the training curve, resulting in an optimal balance between performance efficiency and precision.

The remainder of this paper is organized as follows. Section 3 discusses the transfer analysis and the specifics of the proposed tracking algorithm. Section 4 provides the experimental results and their analysis. Concluding remarks are made in Section 5.

2. Related work

Traditional anti-drone defense systems that depend on spectrum and radar detection are typically expensive and inflexible, making them insufficient to meet the high frequency requirements of anti-UAV tracking. Recently, several datasets, including 'A dataset for small infrared moving target detection under clutter background' [33], 'A dataset for infrared detection and tracking of dim-small aircraft targets under ground/air background' [34], and 'Anti-UAV410' [35], have been presented, aiding in tracking UAVs using visual information.

The two primary DL approaches [36–41] in visual tracking are Siamese networks and transformer models. The fully convolutional Siamese [42] method employs a twin network with shared weights to obtain feature representations, while this approach is sensitive to target deformation. Consequently, the Siamese Region Proposal Network (SiamRPN) [23] and its enhanced version SiamRPN++ [36] are introduced. However, the pooling operation on DL trackers can lead

to a reduction in detailed target information. To mitigate inaccuracies within the infrared anti-UAV drone dataset, the Siamese drone tracker [35] incorporates a dual-semantic extraction method to emphasize the target's semantic significance. The SiamIST [25] variant of the SiamRPN model is proficient at detecting targets within infrared imagery. However, its performance is limited due to a complex anchor mechanism. The successful integration of transformers [37,43] within the visual domain has seamlessly combined them with Siamese architectures. The TransT [37] method integrates an improved effective attention mechanism to replace the conventional correlation process in Siamese. Moreover, the SwinTrack [43] leverages transformers to capture feature representations, which improves long-range modeling abilities for effective feature extraction and integration.

In summary, the DCF algorithm has slowly retreated from the tracking phase, while DL algorithms are continuously emerging. However, the advancement of Siamese trackers seems to be reaching a standstill, leading researchers to explore the use of emerging technologies. Here, we are committed to improving the transformer framework for small UAV tracking.

3. The proposed transformer-based IST algorithm

3.1. Analysis of the transformer track

The TransT reformulates the tracking challenge as a one-time detection issue by employing a pre-trained network [37]. Fig. 1 illustrates that the main network is responsible for extracting features from the template and searching images. Additionally, these features are enhanced and combined using a feature fusion network that incorporates ECA and CFA modules.

Feature fusion. For each input X , the ECA module is used for feature extraction, and the output X_E is formulated as

$$X_E = X + \mathbf{M}(X + P_x, X + P_x, X) \quad (1)$$

where P_x represents the spatial position encoding and $\mathbf{M}$ symbolizes the multi-head attention mechanism.

The template area that refers to the query and the search area (used as key and value for attention fusion) are fused through the CFA module, whose output of CFA X_{CF} is formulated as

$$X_{CF} = X_C + \max(0, X_C W_1 + b_1) \times W_2 + b_2 \quad (2)$$

where $\max$ denotes the maximum function. W_1, W_2 represent the weight matrices, and b denotes the basis vectors. According to (1), each output X_C referring to the attention in the ECA model is formulated as

$$X_C = X_{E1} + \mathbf{M}(X_{E1} + P_{E1}, X_{E2} + P_{E2}, X_{E2}) \quad (3)$$

where X_{E1} and X_{E2} are the two CFA inputs referring to the template and the search areas. P_{E1} and P_{E2} denote the corresponding spatial position encoding. Finally, the two CFA outputs are fused to obtain X_{EC} .

The loss of TransT. Following this, the total loss consists of a classification loss and a regression loss (see Fig. 1). In the classification task, the standard binary cross-entropy loss L_c is calculated as

$$L_c = - \sum_j [y_j \log(p_j) + (1 - y_j) \log(1 - p_j)] \quad (4)$$

where y_j represents the ground-truth label for the j th sample, with $y_j = 1$ indicating the foreground, and p_j is the probability that the learned model predicts the foreground.

Furthermore, the regression loss L_r^+ encompasses the discrepancies between the normalized ground-truth bounding boxes and the predicted ones, calculated by employing the L1 loss L_1 and the generalized intersection over union loss L_G .

$$L_r^+ = \lambda_G L_G + \lambda_1 L_1 \quad (5)$$

where the regularization parameters $\lambda_G = 2$ and $\lambda_1 = 5$ are determined by experimentation.

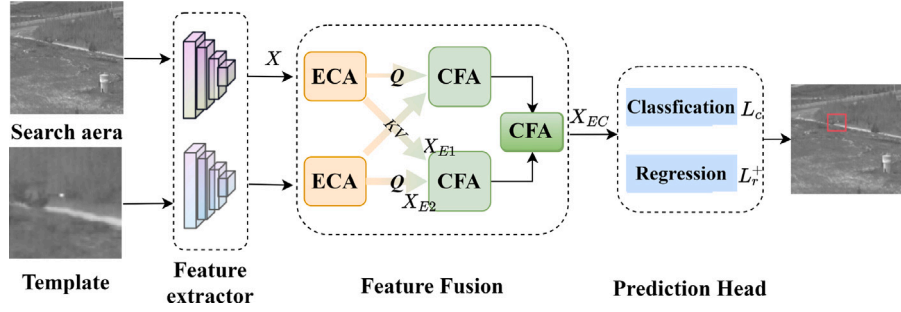


Fig. 1. The framework of TransT.

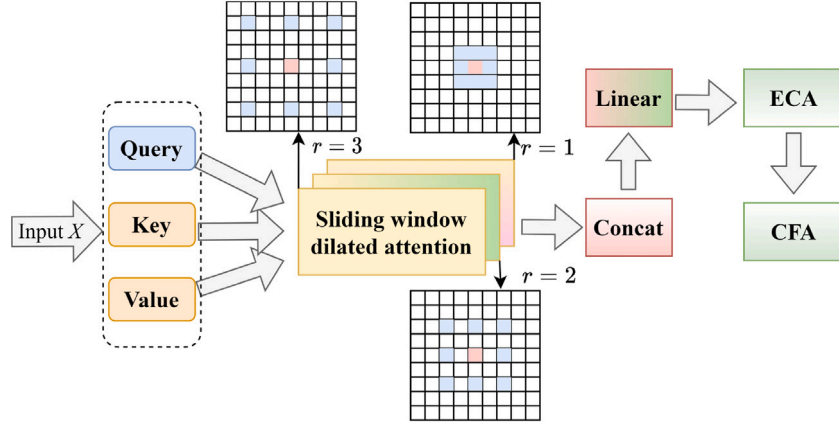


Fig. 2. The MSDA model.

Migration analysis. The TransT method above performs well on visual area targets. However, it is difficult to obtain effective features from a small target, and the complex background makes it a challenge for DL methods without effective training. Therefore, it is necessary to improve the performance of the backbone, which can handle the migration analysis of the IST methods. In addition, the output distribution of the softmax function is easily uneven, affecting the model learning process and even causing the collapse of attention entropy [44].

3.2. The proposed TransIST method

3.2.1. Multi-scale dilated attention

To increase the effectiveness of the TransT approach referring to complex backgrounds, multi-scale dilated attention [45] is implemented. The MSDA technique improves the model's capability to identify small targets by integrating features at different scales. Initially, MSDA uses Sliding Window Dilated Attention (SWDA) to calculate self-attention in localized areas (Fig. 2). Subsequently, the feature aggregation model is utilized to effectively combine data gathered from multiple scales, yielding robust features.

Additionally, the SWDA establishes a window that concentrates on each query point on the feature map. After this, keys and values are selected, which contributes to a decrease in computational load. The SWDA output X is represented as

$$X = \mathbf{L}(\mathbf{C}(\mathbf{F}(Q_i, K_i, V_i, r_i))) \quad (6)$$

where $\mathbf{L}$, $\mathbf{C}$, and $\mathbf{F}$ are feature aggregation, concat, and SWDA operations, respectively. The dilation rate r_i denotes the i th head. Q_i , K_i , and V_i represent the segments of the feature map corresponding to the i th head with the dilation rate r_i .

With SWDA operation, the feature map is segmented into various heads with distinct dilation rates. For small target tracking, the dilation rates should not be set too large. Typically, the size of the kernel is

configured to be 3×3 , with dilation rates of $r = 1, 2, 3$. These rates correspond to receptive fields measuring 3×3 , 5×5 , and 7×7 across various heads. Specifically, the component x_{ij} referring to X is given by

$$x_{ij} = \mathbf{S} \left(\frac{q_{ij} K_r^T}{\sqrt{d_K}} \right) V_r \quad (7)$$

where $\mathbf{S}$ is the softmax operation, and q_{ij} denotes the query at position (i, j) . The variable K_r denotes the keys selected from K with dimension d_K , and V_r refers to the values from V .

According to the above, MSDA not only captures key features at different scales but also enhances the representation of the target by aggregating features. Furthermore, an ablation study is presented in Section 4.4, demonstrating the effectiveness of MSDA in the IST task.

3.2.2. Side window filtering

The previously mentioned MSDA improves the representation of small targets. Despite the enhancements made by MSDA, the task of tracking small targets that are partially obscured by background edges remains a significant challenge. As a result, side window filtering is used to tackle this problem. Specifically, for a region of interest v , the outcome after filtering is denoted by $v^{l, \rho, \theta}$.

$$v^{l, \rho, \theta} = \mathbf{G}(v, l, \rho, \theta) \quad (8)$$

where $\mathbf{G}$ represents the SWF function, l is the radius of the filtered window, ρ signifies the pixel's position at the edge or corner of the window, and θ denotes the window's orientation.

Furthermore, the best result R_v is achieved by reducing the distance between the input v and the output $v^{l, \rho, \theta}$.

$$R_v = \arg \min_{v^{l, \rho, \theta}} \|v - v^{l, \rho, \theta}\|_2^2 \quad (9)$$

where $\|\cdot\|_2$ represents the L2 norm, and the target's information v is given by $(2 * v - R_v)$.

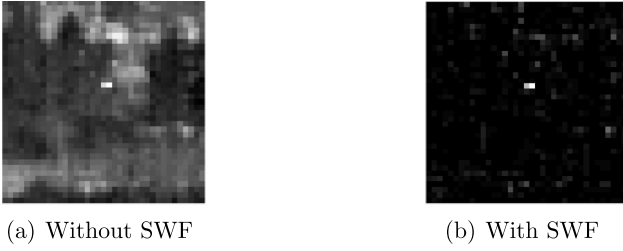


Fig. 3. The effect of SWF referring to a small target.

Following this, the ability of the SWF model to preserve edges is confirmed through tests on infrared images. In Fig. 3(a), the experimental findings demonstrate that the intricate background can be minimized while preserving the high-frequency details of the target. In essence, the model effectively distinguishes between target and edge details, which presents difficulties for conventional filtering techniques.

3.2.3. Exponential moving average learning

Using the MSDA and SWF described previously, the information regarding the target is amplified while the background noise is minimized. However, the model tends to overfit when dealing with small targets. Against this backdrop, the exponential moving average method is adopted to optimize the parameter variation of the IST task. To obtain the state of a target, the core of the EMA model lies in calculating the weighted average of historical information, which is shown in Fig. 4. In particular, the EMA method, which considers the temporal information of the target, provides a stable tracking result in the IST task, effectively mitigating the impact of the dynamic environment.

Specifically, the update mechanism is defined as

$$\theta_E^{(t)} = \alpha \cdot \theta_E^{(t-1)} + (1 - \alpha) \cdot \theta_t \quad (10)$$

where θ_t and θ_E represent current and historical parameters, respectively. t denotes the training epoch number and α controls the degree of blending between historical and current parameters.

The steps and flow chart of the proposed algorithm are shown in Algorithm 1 and Fig. 5. The results and analysis of the experimental simulations will be presented later.

Algorithm 1 The proposed TransIST algorithm

- 1: Process NIR images with the SWF.
 - a. For the initial frame, determine the sampling area labeled as z .
 - b. Compute the SWF result $R(z)$ as (9), and improve the small target using $(2z - R_z)$.
 - 2: Extract features employing ResNet with the MSDA model in (6).
 - 3: Integrate features utilizing the ECA and CFA models as (1)-(3).
 - 4: Obtain classification and regression outcomes using the prediction head.
 - 5: Update the state of the target.
-

4. Experimental results and analysis

The experimental setup is described in Section 4.1, which includes the data sets and experimental details. Sections 4.2 and 4.3 provide a detailed analysis of several existing state-of-the-art IST methods, which helps to discuss the advantages and limitations of the proposed algorithm. Furthermore, the ablation study is conducted in Section 4.4 to demonstrate the effectiveness of each component in the TransIST method.

4.1. Experiment setup

To evaluate the performance of the model, public data sets of small NIR targets [33,34] are used for training. The training data set² is derived from [33,46], which include various scenes with complex backgrounds and targets of 3×3 pixels. Moreover, the test data set³ is obtained from [34], featuring real drones against complex backgrounds. What needs illustration is that the first four video sections were excluded, due to excessively large or numerous targets.

For small targets, the size of the tracking boxes is represented by 10×10 pixels and the search area is set to 32×32 . Taking into account the test efficiency and the target scale, the window radius l in SWF is set to 1. The decay rate of the EMA method is set to 0.001. The number of training epochs is 125. The comparison algorithms contain DCF and DL based methods, including Infrared Patch Image (IPI) [47], KCFCF [13], SiamRPN [23], STCGIF [48], STRCF [18], STRCFD and STRCF2 (the STRCFD method without detection) [20], SiamIST [25], TransT [37], ET-tracker [26], and Tomp [27]. The experiments were carried out on PyCharm with NVIDIA 2080TI GPU. Codes can be requested by email.⁴

4.2. Qualitative comparison

The experimental results of six representative NIR videos are depicted in Figs. 6–11. In Fig. 6, the target navigates through a dense vegetative background, encountering abrupt enhancements in grayscale features. The STRCFD model erroneously assumes similar thermal object targets, leading to a complete migration. KCFCF, STRCF, transt and ET-tracker substantially lose the target in frame 73. The Tomp model also loses the target after strong background changes in frame 113. However, the SiamIST and the proposed TransIST perform well, overcoming the difficulties of strong edges.

Fig. 7 depicts a dim target located between a mountain and the sky. In particular, TransT, SiamRPN STRCFD and KCFCF did not initially detect the target. STCGIF, IPI, and SiamIST are not stable throughout the video. Furthermore, the suggested model demonstrates strong performance in identifying features of small targets, consistently preserving target tracking throughout the sequence.

In Fig. 8, a distinct region with characteristics similar to those of the target area is observed in frame 111. All DCF-based algorithms (KCFCF, STCGIF, STRCF, and STRCFD) together with the TransT algorithm commonly identify the building edge as a target. Nevertheless, the TransIST algorithm effectively differentiates the small target from the complex background with minimal loss.

In Fig. 9, the tracking task is influenced by abundant trees with low resolution. KCFCF and IPI algorithms do not handle strong noise in frame 69, resulting in significant tracking drift. Crucially, the STRCFD and TransT algorithms exhibit instability in target tracking, which stems from an inadequate feature extraction model. By accounting for multi-scale differences between targets and noise, the proposed algorithm successfully tracks a small target.

In Fig. 10, the target moves fast at certain times. STRCF, STRCF2, and KCFCF algorithms acquire a false target from beginning to end. SiamRPN and SiamIST drift into the background in frame 228, indicating instability when the edge appears. However, the proposed algorithm handles strong edges well and tracks the target stably.

In Fig. 11, the strong edge occurs around the mountain area. In detail, the tracking errors of the STRCF and KCFCF algorithms are significant in most frames, with poor edge handling. The two siamese-based algorithms (SiamRPN and SiamIST) achieve good tracking results. The proposed algorithm improves the ability to extract features, which helps in effective small target tracking.

² Available at: <https://www.scidb.cn/en/detail?dataSetId=808025946870251520>.

³ Available at: <https://www.scidb.cn/detail?dataSetId=720626420933459968>.

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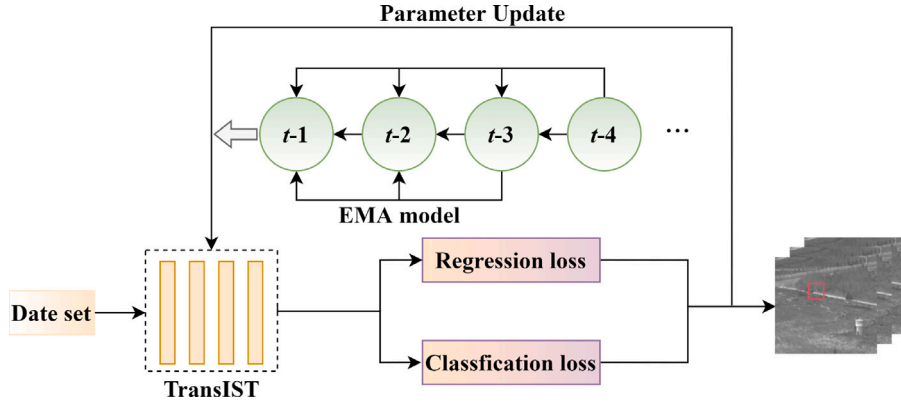


Fig. 4. The schematic diagram of EMA.

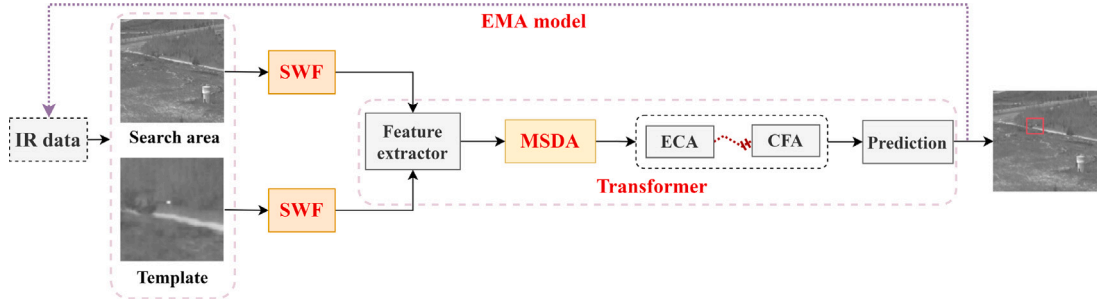


Fig. 5. The flow chart of the proposed TransIST algorithm.

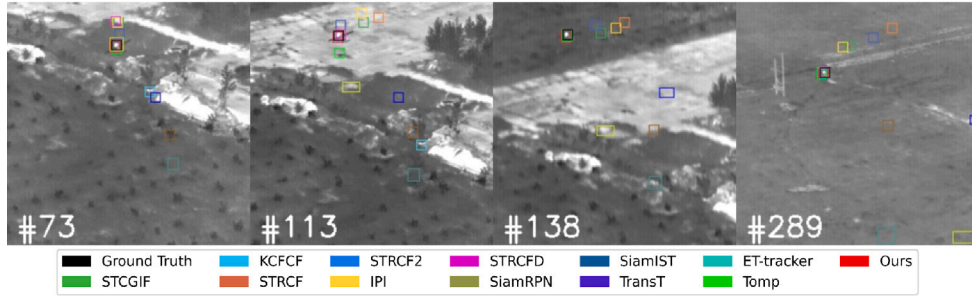


Fig. 6. Qualitative results over Video 9 (Optimal viewing on color screens).

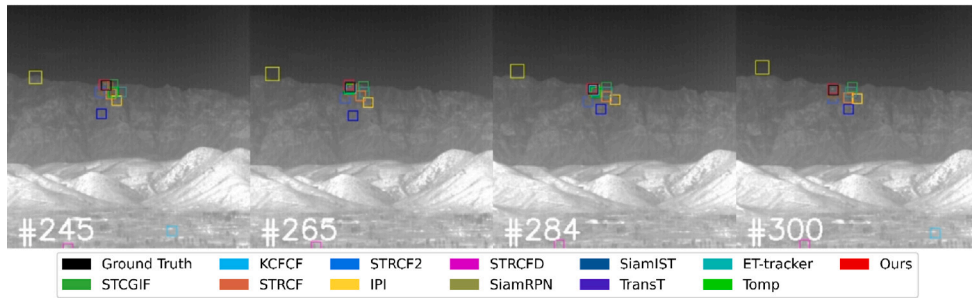


Fig. 7. Qualitative results over Video 10 (Optimal viewing on color screens).

4.3. Quantitative comparison

In this section, we evaluate the performance using the precision plot and the success plot [49], as shown in Figs. 12 and 13. Precision is described as the ratio of frames in which the Center Location Error (CLE) $d_{CLE} = 5$ for the predicted bounding box compared to the ground truth falls below a predefined limit. The success rate is defined as

the percentage of frames in which the overlap between the predicted bounding box and the ground truth exceeds a specified threshold. Furthermore, the Area Under the Curve (AUC) for the success plot signifies the area beneath the performance curve in relation to the horizontal axis.

As shown in Fig. 12, each pertinent algorithm is tested on an NIR data set that includes 300 frames per video. It is evident that the

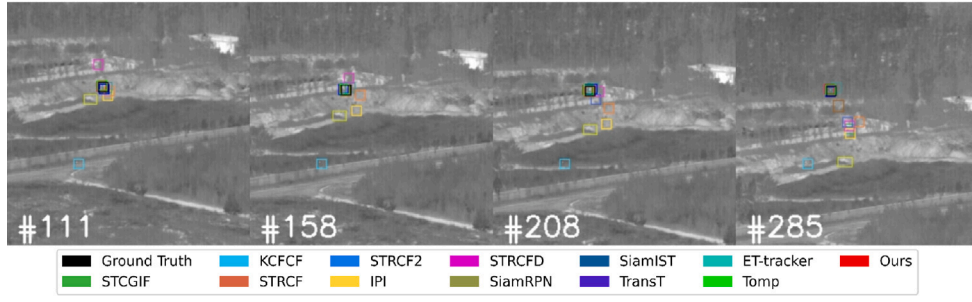


Fig. 8. Qualitative results over Video 11 (Optimal viewing on color screens).

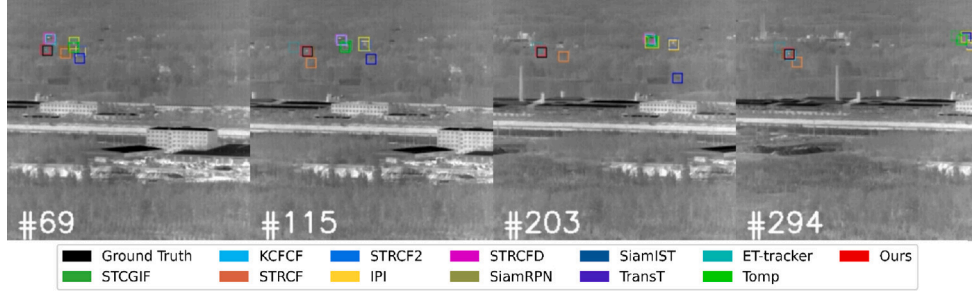


Fig. 9. Qualitative results over Video 15 (Optimal viewing on color screens).

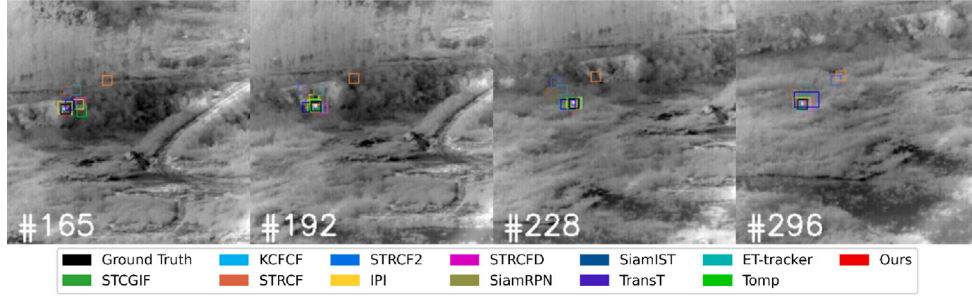


Fig. 10. Qualitative results over Video 18 (Optimal viewing on color screens).

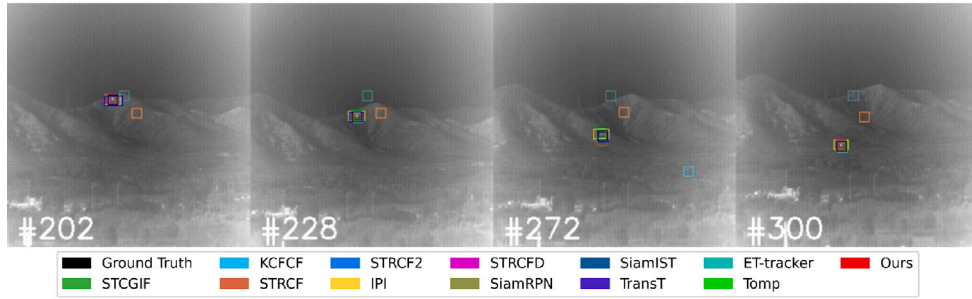


Fig. 11. Qualitative results over Video 20 (Optimal viewing on color screens).

algorithm outperforms other tracking algorithms in both precision and success rate. Moreover, the ablation study conducted on the complete NIR data set validates the robustness of the TransIST algorithm. Subsequently, Table 1 shows a performance comparison for all NIR videos, including precision, AUC of the success plot, and Frames per Second (FPS).

It can be seen that the STRCFD attains a high tracking precision with the second highest AUC value (0.612). While STRCF attains an impressive FPS rate of 145, it struggles to effectively track small targets, primarily because it handles strong edges inadequately. Additionally, the baseline algorithms outperform the proposed method in terms of

FPS (44 compared to 30) because of the SWF module. By excluding the SWF module, the method's frame rate can increase to 40 FPS. However, the TransIST algorithm excels with an overall precision of 0.881 and an AUC of 0.660, showcasing enhanced stability of the tracking box.

To sum up, the proposed algorithm exhibits robust performance, as evidenced by visual displays and excellent evaluation metrics.

4.4. The ablation study

To assess the performance of the suggested algorithm against the baseline, an ablation study is conducted on the transformer track.

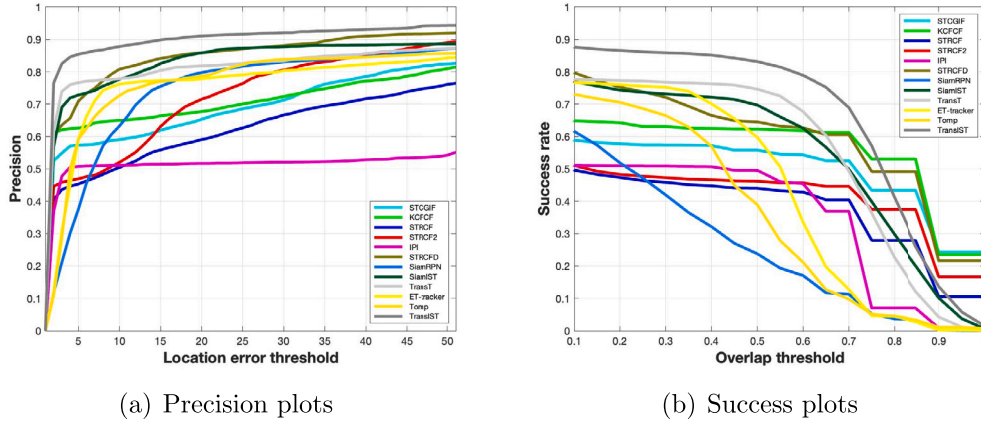


Fig. 12. The performance comparison.

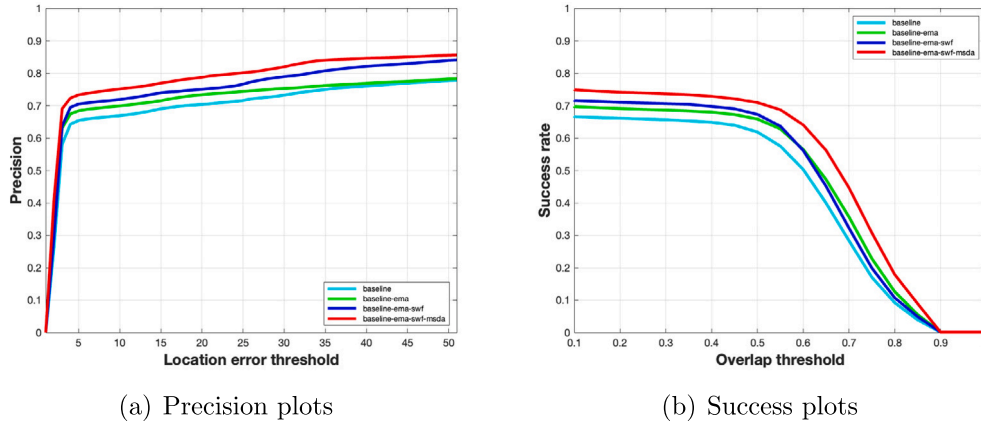


Fig. 13. The ablation comparison.

Table 1

Comparative analysis of algorithms (Precision at $d_{CLE} = 10$ and FPS).

Algorithm	AUC	Overall precision ($d_{CLE} = 10$)	Overall FPS
IPI [47]	0.371	0.511	1
KCF [13]	0.584	0.651	69
STCGIF [48]	0.528	0.594	50
STRCF [18]	0.404	0.514	145
STRCF2	0.444	0.535	50
SiamRPN [23]	0.275	0.662	62
SiamIST [25]	0.549	0.784	30
STRCFD [20]	0.612	0.814	43
TransT [37]	0.572	0.767	44
ET-tracker [26]	0.439	0.765	47
Tomp [27]	0.371	0.741	45
Ours	0.660	0.881	30

Table 2

The ablation study concerning SWF, EMA, and MSDA (with precision $d_{CLE} = 5$ and not confined to 300 frames).

Algorithm	AUC	Precision ($d_{CLE} = 5$)	Overall FPS
baseline	0.410	0.655	44.4
baseline-EMA	0.442	0.685	44.4
baseline-EMA-SWF	0.446	0.706	31.3
baseline-EMA-SWF-MSDA	0.489	0.735	29.8

SWF, MSDA, EMA denote side window filtering, multi-scale dilated attention and exponential moving average.

These studies are carried out under stricter conditions using precision ($d_{CLE} = 5$) and throughout the data set (not limited to 300 frames), incorporating SWF, EMA and MSDA.

As demonstrated in Table 2, the baseline obtained a respectable precision of 0.655, signifying an above average performance. Moreover, incorporating SWF, EMA, and MSDA each enhanced the tracking performance. Firstly, the total precision of the baseline-EMA is 0.685, which surpasses the precision of the baseline algorithm (0.655). The combined precision of baseline-EMA-SWF is 0.706, which is higher than that of the baseline-EMA model. Among the models, baseline-SWF-EMA-MSDA achieves the highest precision of 0.735, highlighting the efficacy of the MSDA model.

5. Conclusions

An improved transformer algorithm has been designed to track small targets in NIR videos. The MSDA effectively obtains multi-scale semantic information of small targets, thereby enhancing tracking performance. High-quality training sets from public data sets play a key role in the generalization of the proposed tracker. Additionally, exponential moving average learning allows TransIST to perform well in training. Moreover, the use of side window filtering helps reduce the adverse effects caused by background edges. Significantly, testing results have demonstrated that the TransIST method attains a total accuracy of 0.881, providing a firm foundation for using transformers in the tracking of small targets.

Even though the proposed TransIST demonstrates promise, it faces challenges related to occlusion and operational efficiency. These challenges highlight the need for further research in these areas. Future work will focus on improving the feature extraction techniques for UAVs, with an emphasis on smart city management.

CRedit authorship contribution statement

Kun Qian: Writing – review & editing, Writing – original draft, Supervision, Resources, Methodology, Data curation, Conceptualization. **Dingwei Zhu:** Writing – original draft, Visualization, Software, Investigation, Data curation. **Yutong Wu:** Writing – review & editing, Visualization, Software. **Jian Shen:** Writing – review & editing, Visualization, Methodology. **Shoujin Zhang:** Writing – review & editing, Methodology, Investigation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work is partially supported by the National Natural Science Foundation of China (62076110), and the Natural Science Foundation of Jiangsu Province, China (BK20181341).

Data availability

Data will be made available on request.

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